

# Notes on the Modules

What are the major understandings we want teachers to have about each module? This includes both content knowledge and an understanding of the motivation behind the module.

## Unit 1

### Module 1.2: Autonomous Robots

- Motivation: kids like robots; robot videos are cool; robots are a good way to introduce AI concepts such as perception and reasoning. So let's start the course off with some fun stuff like dancing robots.
- Major ideas:
  - Autonomy: we're only interested in robots that act on their own. We don't care about remote-controlled robots, or robots that just blindly repeat an action without having to sense anything or make any decisions. Robots that make their own decisions are called **autonomous**.
  - Robots come in many different shapes, but they're all doing the same thing: navigating through the world and interacting with it. Three questions every autonomous robot must constantly ask itself:
    - Where am I?
    - Where do I want to be?
    - How can I stay safe?
  - Self-driving cars are robots. The Roomba vacuum is also a robot. They're doing the same thing: moving through the world and trying not to hit stuff.
  - Autonomous robots must maintain a **representation** (e.g., a map) of the world so they know where stuff is and can figure out where they are. This requires **perception** to sense the world. All autonomous robots have some type of perception.
- What students should be able to do:
  - Explain what autonomy means.
  - Given a robot, explain what it does and what kinds of decisions it makes.
  - Decide whether a device is a robot or not. Roombas are robots. Your grandmother's antique Electrolux vacuum is not a robot. Why not?

### Module 1.3: Self-Driving Vehicles

- Motivation:
  - Further develop our understanding of autonomous robots.
  - Self-driving cars are all over the news, so this subject is topical.

- Many cars have driver-assist features such as lane departure warnings that are powered by computer vision, so students might actually be experiencing these technologies in their own lives. We want them to realize that these are built on AI.
- Major ideas about how self-driving cars work:
  - Sensing: learn about different types of sensors such as cameras, radar, lidar, and GPS. How do they work? What are their strengths and weaknesses?
  - Reasoning: what kinds of things do self-driving cars need to figure out? Route planning, maneuver planning (e.g., safe lane changes), obstacle avoidance, guessing the intentions of other vehicles; figuring out traffic signs.
- Societal impacts of self-driving cars:
  - Do people trust them too much when they're not yet safe to rely on?
  - Who is liable in case of an accident?
  - How will they affect employment?
- What students should be able to do:
  - Explain the sensors on a self-driving car: what they do and where they are located.
  - Explain the decisions a self-driving car makes, or alternatively, explain the subsystems that make up a self-driving car.

## Module 1.4: Perception

- Motivation: the goal for this module is to get students thinking about how computers perceive things, i.e., how they extract meaning from raw sensor signals such as camera images. It's a complicated process which we call the "video abstraction pipeline" that goes from image pixels to recognizing an object, a face, or an entire scene. Explaining how self-driving cars understand the road scene is too complicated, so instead we explore face detection, because it's easier to explain and there are lots of good resources we can draw on.

Face detection is **not** the real goal of the module. Face detection is just an illustrative problem to get students thinking about perception.

- Major concepts:
  - Images are 2D arrays of pixels, which can either be grayscale values or RGB values (for color).
  - Feature extraction finds structure in images. Edge detection is the example we'll focus on, but feature extraction has many forms. Detecting eyes and jawlines, or fitting a triangular mesh to a face, are forms of feature extraction that are built on top of simpler things like edge detection.
  - Perception takes place in stages, starting with the raw signal and proceeding through several stages of feature extraction to end up with a high-level interpretation of the image, e.g., "Oh, that's Betia and she's smiling."
  - "Features" are a major concept in the course and will reappear when we talk about decision trees, machine learning algorithms, and semantic feature spaces.

- What students should be able to do:
  - Explain how some part of perception works, e.g., explain an edge detector, or explain how face templates are aligned with a face image (see the Snapchat video).
  - Build their own Snapchat filter in Scratch.

## Module 1.5: Route Finding

- Motivation: the real purpose of this module is to introduce students to breadth-first search, tree and graph structures, and (implicitly) the notion of a state space. We will go into these concepts in greater depth later in the course when we talk about reasoning and problem solving. We're able to introduce them here because we're focusing on a problem related to autonomous robots: finding a route to a destination.
- Major concepts:
  - Graphs and trees
  - How to encode information as a graph
  - Breadth-first search by graph coloring
  - Breadth-first search by constructing a search tree
  - Reasoning algorithms (breadth-first search is one)
- What students should be able to do: solve route-finding problems using two methods: (1) by coloring the graph, or (2) by constructing a search tree.
  - Graph coloring is more fun, but some types of search problems can't be solved that way; they require us to construct a search tree.
  - This is why we want students to learn both methods. They'll use the search tree method later in the course when we look at AI algorithms for game playing and puzzle solving.

## Module 1.6: Case Study

This case study focuses on people sleeping in their Teslas.

- Motivation: help students think about the impacts of technology, both good and bad, and how new technologies can pose challenges for the law or for social policy.
- Major concepts:
  - Fully autonomous self-driving cars (none yet) vs. somewhat-autonomous cars that still require human monitoring (e.g., Tesla Autopilot).
  - Legal responsibilities of drivers.
  - Legal responsibilities of the car manufacturer.
  - Threat or boon? Self-driving cars have killed a few people, but in general they are better at avoiding accidents than human drivers. So what should our policy be for approving their use? When will they be "good enough"? Discuss among yourselves.
- What students should be able to do:
  - Identify the safety/social issues with self-driving cars.

- Describe the current state of autonomous driving technology.
- Debate what to do about people sleeping in their Teslas.

## Module 1.7: Mini-Project

A variety of mini-projects are possible. The Snapchat filter exercise in Scratch is one good option. Another is to use Calypso to build a little environment with walls and obstacles and explore the robot's ability to do route finding through this environment. Another option is to use Teachable Machine to learn to recognize traffic signs such as stop signs and pedestrian crossing signs.

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## Unit 2: Computers and Language

### Module 2.1: Introduction to the Unit

- **Motivation:** get students to think about the various ways they use language to interact with computers. This includes both spoken language (speech recognition and generation) and text-based interactions such as search queries and chatbots. We will examine all these topics in greater detail in the modules that follow.
- **Technology covered** includes smart assistants such as Siri or Alexa, and text generation applications such as chatbots and story generators.
- This is an abbreviated module that may not take up an entire class period. An earlier draft included material on the nature of language that has been removed to save time, but some of that material is covered in later units where it is relevant.
- **Students should be able to:**
  - List various ways in which they or people around them use language to interact with computers in their daily lives.
  - Distinguish concepts such as speech recognition, speech generation, and question answering.

### Module 2.2: Speech Recognition

- **Motivation:** Give students an end-to-end understanding of how speech recognition works, from the microphone all the way to meaning.
- **Major concepts:**
  - The nature of sound (variations in air pressure)
  - Waveform and spectrogram representations of sound
  - The human vocal tract: how people produce sounds
  - The audio pipeline: extracting phonemes from speech spectrograms and then assembling them into morphemes, words, phrases, and sentences.
  - Ambiguity of the speech signal. Knowledge is required at each stage of the pipeline in order to select the most plausible hypotheses.
- **Supporting software:** online demos that students can experiment with.
  - Oscilloscope demo shows waveforms. See the *Waveforms and Spectrograms* activity resource guide.
  - Spectrogram demo shows spectrograms. See the *Waveforms and Spectrograms* activity resource guide.
  - Vocal tract demo allows students to manipulate a model of the vocal tract and hear the sounds it produces. Works best on a device with a touch screen. Reinforces teaching about the anatomy of the vocal tract.
  - SpeechDemo demonstrates how the Google Speech API considers multiple hypotheses when recognizing spoken language. See the *SpeechDemo* activity resource guide.

- **Student worksheets:** see Amber's worksheets on recognizing spectrograms and testing SpeechDemo on homophones.
- **Students should be able to:**
  - Explain what waveforms and spectrograms are.
  - Explain the components of the human vocal tract.
  - Give examples of how knowledge is used to disambiguate speech, e.g., disambiguating homophones based on context.

## Module 2.3: Understanding Search Queries

- This module is currently labeled "How Do Intelligent Assistants Understand and Answer Questions?", but that title is no longer appropriate as the module now focuses on search queries such as one might give to Google, not interaction with assistants like Siri or Alexa. A better title might be "**How Do Computers Understand Search Queries?**"
- **Motivation:** this module introduces students to natural language processing by focusing on the problem of extracting meaning from words to understand a search query. Meaning extraction shows up in many other places, such as machine translation, but search queries are a good place to start.
- **Major concepts:**
  - What are the different kinds of queries Google can understand and respond to?
  - Parts of speech: nouns, verbs, adjectives, prepositions, etc.
  - Parsing: determining the grammatical structure of a sentence, which tells us how words are associated with other words.
  - Semantics: extracting the meaning of a sentence from the parse tree. This is addressed by showing how an English query can be turned into a bit of computer language that looks like a database query. Running the query against Google's "knowledge graph" (a kind of database) finds the answer.
- This is an abbreviated lesson that will probably take less than one class period.
- **Students will be able to:**
  - Explain what a parse tree is, and list some parts of speech.
  - Describe different types of queries that Google can answer.
- This module has been shortened a bit. An earlier draft had students experiment with the Berkeley Neural Parser (now we just show one figure generated by the parser), explore cases where a sentence might have more than one possible parse tree, and discuss different types of ambiguity that can occur in sentences, e.g., prepositional phrase attachment, or pronoun reference resolution.

## Module 2.4: Word Embeddings

- **Motivation:** The reason to teach word embeddings is that they introduce students to a powerful concept that underlies much of modern AI: numerical vector representations. The previous module on search queries looked at symbolic representations such as parse trees and knowledge graphs. Numerical vector representations are the alternative

to symbolic representations. They are what's inside neural networks, including neural networks that perform important language processing tasks such as machine translation.

- **Secondary motivation:** Word embeddings are a type of feature vector representation, so teaching them lays the groundwork for considering other examples of feature vector representations in unit 3 on machine learning.
- **Major concepts:**
  - Semantic dimensions, e.g., age, gender, royalty
  - Semantic feature space
  - Words as points in a semantic feature space
  - The coordinates of a word in feature space yield a numerical representation of the word.
  - Semantic similarity corresponds to closeness in feature space
  - Solving word analogies using vector arithmetic
- **Supporting software:**
  - Google's Semantris game raises the question of how Semantris knows that two words are related to each other. Semantris is powered by word embeddings.
  - The Word Embedding Demo allows students to graphically explore 300-dimensional word embeddings projected into a 3D space. It can also be used to solve analogies by vector arithmetic.
- **Unplugged activity:** Will and Amber's semantic feature space unplugged activity.
- **Students will be able to:**
  - Explain the notion of semantic feature space
  - Given a semantic feature space, determine where a word should be located in that space
  - Demonstrate how to solve analogy problems using vector arithmetic

## Module 2.5: Machine Translation

- This module has been made optional due to lack of time, and because it may not be engaging for middle school students.
- **Motivation:** take a closer look at the computer's ability to represent the meaning of language by exploring the problem of machine translation. In order to correctly translate a sentence we must grasp its meaning; simple-minded word-at-a-time translation doesn't work. So how good is Google Translate at capturing meaning?

## Module 2.6: Sentiment Analysis

- Slide deck is adapted from materials prepared by AI4ALL.
- **Motivation:** introduce students to another way of processing language: detecting whether a passage is expressing positive or negative sentiment. Sentiment analysis tools are used by companies to monitor social media in order to detect how the public is reacting to products or events associated with the company.

- The meat of this module is an activity where students use an online sentiment analysis tool to analyze text or song lyrics.
- **Students will be able to:**
  - Explain what sentiment analysis is.
  - Look at a piece of text and point out words or phrases that will likely lead a sentiment analysis program to conclude that the sentiment is positive or negative.

## Module 2.7: Chatbots

- **Motivation:** Introduce students to the possibility of having a conversation with the computer instead of writing one-shot queries. Discuss different types of chatbots to help students appreciate the difference between simple-minded approaches (e.g., keyword scanning) vs. more sophisticated approaches.
- **Major concepts:**
  - Different uses for chatbots, e.g., customer service
  - Different strategies for constructing chatbots, e.g., keyword matching
  - Differences between chatbots and intelligent assistants
  - Limitations of chatbots

## Unit 3: Machine Learning and Automated Decision Making

### Module 3.1: How Do Computers Make Decisions?

- **Motivation:** Making decisions is done by reasoning. This module describes several types of reasoning (classifying, predicting, recommending, and planning) and examines how computers use reasoning to provide services such as spam detection or movie recommendation.
- **Major concepts:**
  - Types of reasoning problems
    - Classification
    - Prediction
    - Recommendation
    - Planning and scheduling
  - Reasoning algorithms (there can be multiple algorithms for a specific type of reasoning)
- **Students will be able to:**
  - List some different types of reasoning computers can do
  - Determine what kind of reasoning is involved in a given computer application

### Module 3.2: How Can A Computer Classify Objects?

- **Motivation:** To show how reasoning is actually performed by a machine, we use classification by decision trees as an example. Decision trees were chosen because they are easy to visualize and understand. Each node of the decision tree examines a feature of the object being classified, so this naturally leads to a discussion of the use of feature vector representations to describe objects.
- **Major concepts:**
  - Decision trees
  - Tree terminology
    - Root node
    - Children of a node
    - Leaf node (or “terminal” node)
  - Features and feature vectors
- **Activity:** construct a decision tree classifier via the Pasta Land activity or Guess the Animal.
- **Students will be able to:**
  - Explain how to use a decision tree to classify an object.
  - Demonstrate how to extend a decision tree to include a new class.

## Module 3.3: How Can We Train A Computer To Make Decisions?

- **Motivation:** Rather than construct decision trees by hand, which can be quite complex for realistic problems, it is better to get the computer to construct them for us. It can do that if we have defined a feature set, assembled a collection of training data, and created a feature vector representation for each training example. Then we can use machine learning tools to do the hard part: construct the decision tree. To let students experience what it “feels like” to be a decision tree learning algorithm, we have them learn classes they are not already familiar with, so they can’t use their prior knowledge to leap to the answer. Instead they must systematically examine the data and consider hypotheses, which is what real decision tree learning algorithms do.
- **Major concepts:**
  - The “fishy feature space” used in this exercise
  - Entertaining multiple hypotheses, using data to weed out the wrong ones
  - Automated construction of decision trees
  - Using a feature test to “split” a collection of examples into two subsets
  - The machine learning pipeline
- **Supporting software:**
  - MachineLearningForKids (<https://machinelearningforkids.co.uk/>) offers automated decision tree learning and will also draw the tree and allow you to run test examples through the tree to see how they are classified.
- **Note on two learning algorithms presented:**
  - The first learning algorithm, for predicting reef-dwellers, is a very simple algorithm designed to just give the flavor of machine inference. We consider every possible node (feature test) that could predict reef dwelling, and we eliminate those candidates that are contradicted by the training data. As more data come in, we eliminate more candidates, until only one is left. This algorithm assumes that the problem can be solved with a single feature test, and it’s only appropriate in this special case. For more complex problems we could consider every possible decision tree with two tests (two non-terminal nodes), but the number of possibilities would be too large for students to enumerate by hand.
  - The second learning algorithm is closer to what real machine learning systems use. Instead of data coming in a little bit at a time, we consider the entire dataset from the beginning, and put all the examples into the root node. We then consider a feature test that could split these examples into two sets in a way that helps to separate them by class. Once we’ve chosen a feature test (e.g., pectoral fin shape = “smooth”), we split the data across the two child nodes and repeat the process. If all the data in a child node is in the same class, no further splitting is necessary. Otherwise we choose another feature and use it to split the child. This is the approach that solves the poisonous fish example, but rather than have students hand simulate it, we let MachineLearningForKids do the work for us. Then we do the same thing again with Angelique’s Disease, and find that the machine-generated decision tree might not be as clever as one we generate ourselves. But in the Titanic dataset the situation is so complex that we are better off letting the machine solve it.

- **Activity:** train decision tree classifiers to detect poisonous fish and predict which patients will develop Angelique's disease.
- **Students will be able to:**
  - Explain how decision tree classifiers work. (They should have gotten this from Pasta Land but it's reinforced here.)
  - Show how training examples can be encoded as feature vectors.
  - Use MachineLearningForKids to set up and solve a machine learning problem.

## Module 3.4: How Can We Use Machine Learning To Answer Questions With Datasets?

- **Motivation:** This module takes students deeper into the topic of machine learning in several ways:
  - Use of realistic datasets with hundreds of examples instead of toy datasets with only 10-12 examples.
  - Use of continuous feature values (such as cotton content or pocket height) in addition to discrete categorical features such as brand or style.
  - Going beyond classification problems to include prediction problems, e.g., predicting price.
  - Introducing a new machine learning tool: Code.org's AI Lab
  - A taste of data science using AI Lab's data visualization facilities to plot distributions of feature values using histograms or scatter plots.
  - Going beyond decision trees: AI Lab uses something called a "k nearest neighbor" reasoning algorithm instead of a decision tree, so there is no decision tree diagram to display.
  - Coping with variability in results: because AI Lab randomly splits the dataset into a training set (90%) and a test set (10%), and the datasets are small to begin with, we can get different results when we run the same problem several times. Understanding why this occurs leads to a deeper understanding of how machine learning works.
- **Major concepts:**
  - Training set vs. test set
  - Types of features: categorical, discrete numerical, continuous numerical
  - Prediction (of continuous values) vs. classification (choosing a discrete value)
  - Statistical distribution of feature values (histograms, scatter plots)
- **Activity:** predicting the price of jeans based on various features, using a dataset built in to AI Lab.
- **Students will be able to:**
  - Use AI Lab to explore a dataset.
  - Train a classifier or predictor using AI Lab.
  - Measure the performance of a classifier or predictor on a test set.

- Explain why running the same experiment in AI Lab multiple times may produce different results.

## Module 3.5: Neural Networks

- **Motivation:** Neural networks are a key technology for artificial intelligence: many AI applications rely on neural network reasoners for tasks such as speech recognition, computer vision, and language understanding. Neural networks must be trained using machine learning, because setting their many weights (parameters) manually is too difficult for anything but toy problems. The details of how neural networks learn and reason are too complex for K-12, so this module offers just an introduction, covering some basic terminology and the operation of a single computing unit (neuron). Real-world neural network applications have thousands or even millions of these neurons. But even a single neuron can do quite a bit, as some of the examples illustrate.
- **Major concepts:**
  - Neurons
  - Weights
  - Thresholds
  - Linear threshold unit
  - Layers: input, hidden, output
  - Deep neural networks (many hidden layers)
  - Learning by adjusting weights and thresholds
- **Supporting software:**
  - Neuron Sandbox: <https://www.cs.cmu.edu/~dst/NeuronSandbox>
- **Activities:**
  - “Will this dog bite me?” – hand simulate one-neuron learning.
  - It’s possible to use Neuron Sandbox to solve problems by finding the correct weights or threshold for a linear threshold unit. The current slides will need to be amended to support this